

MELBOURNE AP, AustraliaWMO: **948660**Lat: **37.6656S**Lon: **144.8322E**Elev: **390**StdP: **14.49**Time Zone: **10.00 (AUS)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.1	38.9	27.0	21.2	71.0	30.3	24.7	61.4	35.9	52.2	32.4	51.9	8.1	0	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.9	95.8	64.6	91.2	64.1	86.3	63.3	69.5	82.6	67.6	81.6	66.0	79.6	17.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	97.0	72.9	63.8	89.9	70.9	62.0	84.4	69.7	33.8	82.1	32.3	81.8	30.9	79.7	78.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 31.0	27.4	24.2	DB	32.8	105.1	1.5	4.3	31.8	108.1	30.9	110.6	30.1	113.0	29.0	116.2	(4)
(5)			WB	32.1	73.3	1.4	2.5	31.1	75.1	30.3	76.5	29.5	77.9	28.5	79.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.4	68.4	68.5	64.9	59.1	54.3	50.2	49.1	50.3	53.5	57.1	61.3	64.6	(6)	
	DBStd	8.90	7.44	7.06	6.58	5.51	4.41	3.49	3.26	3.88	4.88	6.15	6.94	7.27	(7)	
	HDD50	166	0	0	0	1	8	39	56	43	15	4	0	0	(8)	
	HDD65	2943	41	38	84	191	333	445	493	456	349	259	157	97	(9)	
	CDD50	3225	572	517	462	275	140	44	28	52	118	225	338	454	(10)	
	CDD65	528	148	135	81	16	1	0	0	0	2	15	45	85	(11)	
	CDH74	6171	1714	1420	845	153	3	0	0	0	19	237	621	1159	(12)	
	CDH80	2788	879	666	335	27	0	0	0	0	2	69	244	566	(13)	
(14) Wind	WSAvg	11.8	11.9	11.3	10.9	10.5	11.2	11.7	12.9	13.0	12.9	12.3	11.7	11.8	(14)	
(15) Precipitation	PrecAvg	21.1	1.7	1.7	1.3	1.8	1.6	1.5	1.4	1.8	1.9	2.2	2.3	1.9	(15)	
	PrecMax	32.3	4.0	7.9	3.4	5.6	6.1	4.1	3.7	3.8	5.0	5.7	6.2	5.5	(16)	
	PrecMin	12.2	0.2	0.0	0.2	0.2	0.3	0.5	0.3	0.6	0.4	0.6	0.7	0.1	(17)	
	PrecStd	4.8	0.9	1.9	0.7	1.3	1.1	0.8	0.8	0.8	1.0	1.3	1.4	1.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	104.3	100.0	96.2	83.7	72.9	63.8	62.5	68.3	76.4	87.4	93.1	100.4	(19)	
		MCWB	66.3	66.4	64.2	59.3	56.1	54.1	52.2	55.0	57.9	59.8	62.5	64.7	(20)	
	2%	DB	96.1	93.3	88.1	78.5	68.1	60.9	59.2	63.7	70.3	80.6	87.3	93.1	(21)	
		MCWB	65.1	64.9	63.1	58.8	55.8	53.2	50.8	52.5	55.5	58.1	61.9	63.0	(22)	
	5%	DB	89.8	88.2	82.7	73.9	64.7	58.9	57.2	60.6	66.5	75.1	81.0	86.1	(23)	
		MCWB	64.8	64.7	62.4	58.2	54.9	51.7	49.7	50.8	54.3	56.9	61.0	62.0	(24)	
	10%	DB	83.4	82.8	78.0	70.1	62.4	56.9	55.4	57.7	63.3	70.0	75.5	80.0	(25)	
		MCWB	64.1	64.4	61.7	57.3	53.8	50.7	48.8	49.4	52.7	55.2	59.6	61.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.4	71.8	68.8	63.7	60.4	57.2	54.4	56.3	60.6	63.6	68.2	69.4	(27)	
		MCDB	84.3	84.0	82.5	75.9	66.1	60.6	59.7	64.6	69.1	76.7	81.7	81.2	(28)	
	2%	WB	69.7	69.2	66.5	61.6	57.8	54.0	52.3	53.7	57.6	60.9	65.3	66.4	(29)	
		MCDB	82.7	82.4	78.9	71.2	64.4	58.7	57.1	62.0	66.7	72.9	78.0	80.5	(30)	
	5%	WB	67.6	67.4	64.9	60.1	56.1	52.7	50.9	51.8	55.5	58.8	63.1	64.6	(31)	
		MCDB	83.2	80.2	76.3	69.1	62.5	57.4	55.4	58.4	64.7	70.7	75.7	79.4	(32)	
	10%	WB	65.6	65.7	63.2	58.5	54.7	51.3	49.8	50.3	53.5	56.7	60.9	62.8	(33)	
		MCDB	80.3	78.5	74.3	67.6	61.0	55.9	54.0	56.2	61.9	67.7	72.7	76.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	21.9	21.1	19.7	17.9	14.2	12.6	12.5	14.6	17.3	19.9	20.6	21.8	(35)	
		MCDBR	34.9	31.6	29.2	24.7	17.7	15.2	14.6	18.2	22.4	29.3	31.6	35.1	(36)	
	5% WB	MCWBR	10.4	10.2	10.4	10.5	9.2	9.1	8.5	9.7	11.2	12.3	11.8	11.3	(37)	
		MCWBR	26.7	25.3	23.1	20.4	15.1	12.5	12.4	16.6	20.1	24.2	25.2	26.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.347	0.351	0.326	0.326	0.309	0.298	0.298	0.311	0.329	0.333	0.345	0.339	(40)	
		taud	2.508	2.504	2.574	2.540	2.573	2.592	2.584	2.531	2.480	2.488	2.480	2.520	(41)	
	Ebn at Noon		313	304	299	279	266	260	267	280	293	306	311	317	(42)	
		Edh at Noon	36	34	30	27	22	20	22	26	32	34	36	36	(43)	
(44) All-Sky Solar Radiation	RadAvg		2279	1988	1590	1102	730	565	630	892	1273	1697	1997	2240	(44)	
	RadStd		146	105	92	66	44	45	38	62	93	138	122	126	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.85	N/A	N/A	+1.49	N/A	N/A	-34	-211	+286	+104	(46)
(47)	Regional (38 neighbors)	+0.92	N/A	N/A	+1.58	N/A	N/A	-31	-225	+313	+108	(47)

Nomenclature: See separate page